

```
wget http://archive.apache.org/dist/flume/1.4.0/apache-flume-1.4.0-bin.tar.gz
```

Extract the file from the Flume tar file, as follows:

```
tar -xvf apache-flume-1.4.0-bin.tar.gz
```

Put the *apache-flume-1.4.0-bin* directory inside the */usr/lib/* directory:

```
sudo mv apache-flume-1.4.0-bin /usr/lib/
```

Download the *flume-sources-1.0-SNAPSHOT.jar* and add it to the Flume class path (the JAR contains the Java classes to pull the tweets and save them into HDFS):

```
sudo mv Downloads/flume-sources-1.0-SNAPSHOT.jar /usr/lib/apache-flume-1.4.0-bin/lib/
```

Check whether the Flume snapshot has moved to the *lib* folder of Apache Flume:

```
ls /usr/lib/apache-flume-1.4.0-bin/lib/flume*
```

Copy *flume-env.sh.template* content to *flume-env.sh*:

```
cd /usr/lib/apache-flume-1.4.0-bin/
sudo cp conf/flume-env.sh.template conf/flume-env.sh
```

Edit *flume-env.sh*:

```
sudo gedit conf/flume-env.sh
```

Set *JAVA_HOME* and *FLUME_CLASSPATH* in the *flume-env.sh*

```
JAVA_HOME = <Path to Java JDK Folder>
```

```
FLUME_CLASSPATH = "/usr/lib/apache-flume-1.4.0-bin/lib/flume-sources-1.0-SNAPSHOT.jar"
```

conf/flume.conf should have all the agents (Flume, memory and HDFS) defined as below.

To configure the Flume agent, type:

```
TwitterAgent.sources = Twitter
TwitterAgent.channels = MemChannel
TwitterAgent.sinks = HDFS
```

```
TwitterAgent.sources.Twitter.type = com.cloudera.flume.source.TwitterSource
TwitterAgent.sources.Twitter.channels = MemChannel
TwitterAgent.sources.Twitter.consumerKey = <consumerKey>
TwitterAgent.sources.Twitter.consumerSecret =
```

```
<consumerSecret>
TwitterAgent.sources.Twitter.accessToken = <accessToken>
TwitterAgent.sources.Twitter.accessTokenSecret = <accessTokenSecret>
```

TwitterAgent.sources.Twitter.keywords = opensource, OSFY, opensourceforyou -> Here you could add multiple keywords into the source engine

```
TwitterAgent.sinks.HDFS.channel = MemChannel
TwitterAgent.sinks.HDFS.type = hdfs
TwitterAgent.sinks.HDFS.hdfs.path = hdfs://localhost:9000/user/flume/tweets/
TwitterAgent.sinks.HDFS.hdfs.fileType = DataStream
TwitterAgent.sinks.HDFS.hdfs.writeFormat = Text
TwitterAgent.sinks.HDFS.hdfs.batchSize = 1000
TwitterAgent.sinks.HDFS.hdfs.rollSize = 0
TwitterAgent.sinks.HDFS.hdfs.rollCount = 10000
```

```
TwitterAgent.channels.MemChannel.type = memory
TwitterAgent.channels.MemChannel.capacity = 10000
TwitterAgent.channels.MemChannel.transactionCapacity = 100
```

The *consumerKey*, *consumerSecret*, *accessToken* and *accessTokenSecret* have to be replaced with those obtained from <https://dev.twitter.com/apps>.

TwitterAgent.sinks.HDFS.hdfs.path should point to the *NameNode* and the location in HDFS where the tweets will go to.

The *TwitterAgent.sources.Twitter.keywords* value can be modified to get the tweets for some other topic like football, movies, etc.

To start Flume, type:

```
bin/flume-ng agent --conf ./conf/ -f conf/flume.conf -Dflume.root.logger=DEBUG,console -n TwitterAgent
```

Apache Flume is extensively used these days due to the reliable transmission (one sender and one receiver) it provides and also due to its recovery features, which rapidly help us in the event of a crash. Also, since this is a part of the Apache Hadoop umbrella, we can get good support and always have good options to utilise all the features this tool provides. 

References

[1] https://en.wikipedia.org/wiki/Big_data

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